Task allocation sheet

Formative 3: Deep Q Learning, Machine Learning Techniques II

Group members: Shima Serein, Gaddiel Irakoze
Environment: `ALE/Pong-v5` (Atari Pong)
Repository: github.com/supserrr/deep-q-learning

Task	Member	Where to find it	When
1. Implementation			
Environment setup: Pong with Atari preprocessing, uv project	**Shima**	`pyproject.toml`, `uv.lock` `make_env` in `train.py`	8 Nov
Training script: CNN Q-network, replay buffer, epsilon-greedy, Bellman update, target network, logs and plots	**Shima**	`train.py`	8 Nov
Evaluation script: greedy play shown in real time	**Shima**	`play.py` `tools/retro_view.py`	8 Nov
Experiment runner and Colab notebook, used by both members	**Shima**	`tools/experiments.py` `notebooks/train_on_colab.ipynb`	8 Nov
2. Hyperparameter experiments			
10 experiments, one hyperparameter changed at a time	**Shima**	`models/shima/exp01` to `exp10` Shima's results file in `results/`	8 Nov
10 experiments, several hyperparameters changed together	**Gaddiel**	`models/gaddiel/exp01` to `exp10` Gaddiel's results file in `results/`	10 Nov
Comparison of both members' experiments	**Gaddiel**	`docs/experiments.md` (Comparison)	10 Nov
800,000-step run of the best combined configuration	**Gaddiel**	Reported in Gaddiel's results file	10 Nov
3. Final agent and documentation			
Final agent: best configuration retrained for 800,000 steps	**Shima**	`dqn_model.pth` `models/shima/best/`	8 Nov
Gameplay recordings of the final agent	**Shima**	`videos/`	8 Nov
README and docs: architecture, results, Colab guide	**Shima**	`README.md` `docs/`	8 Nov
Review and merge of pull requests	**Shima**	Pull requests #1 and #2	12 Nov
4. Presentation (Week 6)			
Introduction: the Pong environment and its goal (1 min)	**Gaddiel**	Slides	Week 6
DQN implementation: Q-network, replay buffer, target network, epsilon-greedy, Bellman loss (2 min)	**Shima**	Slides, `train.py`	Week 6
Own 10 experiments: best and worst, final choice (2 min each)	**Both**	Slides, each member's results file	Week 6
Live gameplay demonstration (2 min)	**Gaddiel**	`uv run play.py`	Week 6
Questions and answers (10 min)	**Both**	Live	Week 6

Dates in sections 1 to 3 are when the work was pushed to the repository (November 2026).